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EXPERIMENT 9

Exp No:9 Experiment to understand Matplotlib library use cases in Data Science through visualization

AIM:
To understand Matplotlib library use cases in Data Science through visualization

DESCRIPTION:
Use Titanic data set to understand the Matplotlib library.

CODE:

```
import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv('/content/Titanic-Dataset.csv')

pd.set_option('display.max_columns', None)

pd.set_option('display.width', None)

display(data)

data.info()

display(data.describe())

display(data.value_counts('Survived'))

sns.countplot(x='Survived', data=data)

plt.show()

dummies = pd.get_dummies(data.Sex)

FinalDataset = pd.concat( [pd.get_dummies(data.Sex), data.iloc[:,[0,1,2,5]]],axis=1)

display(FinalDataset.head())

sns.scatterplot(x='Age',y='Fare', hue='Survived',data=data)
```

```
plt.show()
sns.scatterplot( x='Pclass', y='Fare',hue='Survived',data=data)
plt.show()
sns.pairplot( data[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']],hue='Survived',
height=3)
plt.show()
sns.FacetGrid( data,hue='Survived', height=5).map(sns.histplot, 'Age').add_legend()
plt.show()
sns.FacetGrid( data, hue='Survived', height=5).map(sns.histplot, 'Fare').add_legend()
plt.show()
sns.FacetGrid(data,hue='Survived', height=5).map(sns.histplot, 'Pclass').add_legend()
plt.show()
sns.FacetGrid( data,hue='Survived', height=5).map(sns.histplot, 'SibSp').add_legend()
plt.show()
```

OUTPUT:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradl	female	38.0	1	0	PC 17599	71.2833	C85	C

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
				ey (Florence Briggs Th...								
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

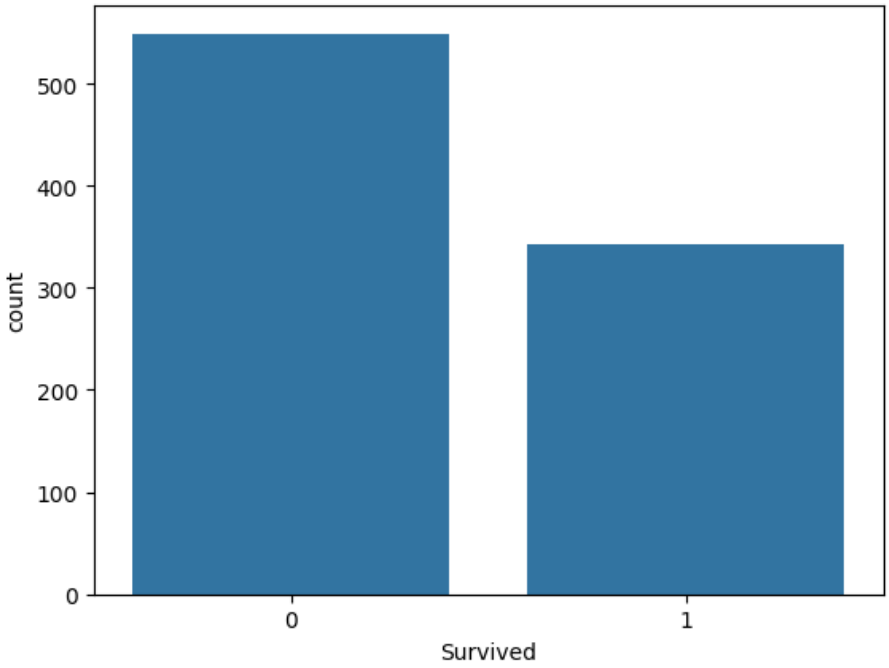
memory usage: 83.7+ KB

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000

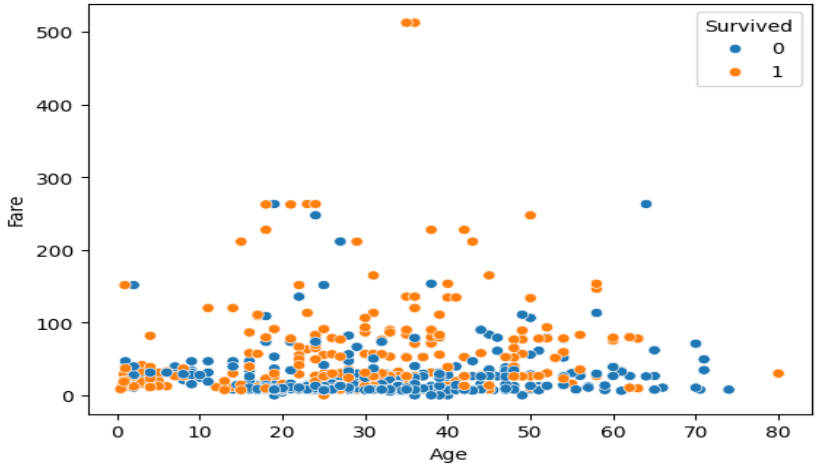
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

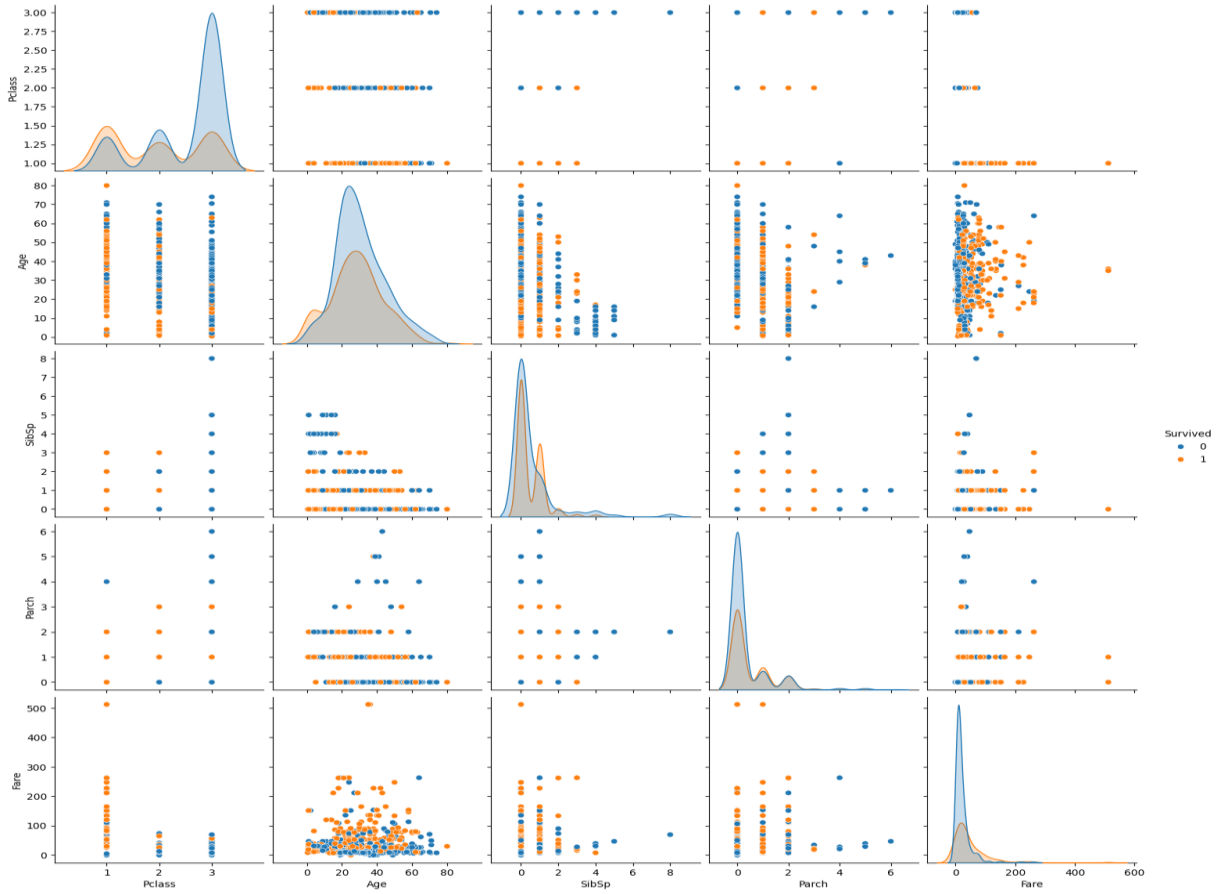
Survived	**count**
0	549
1	342

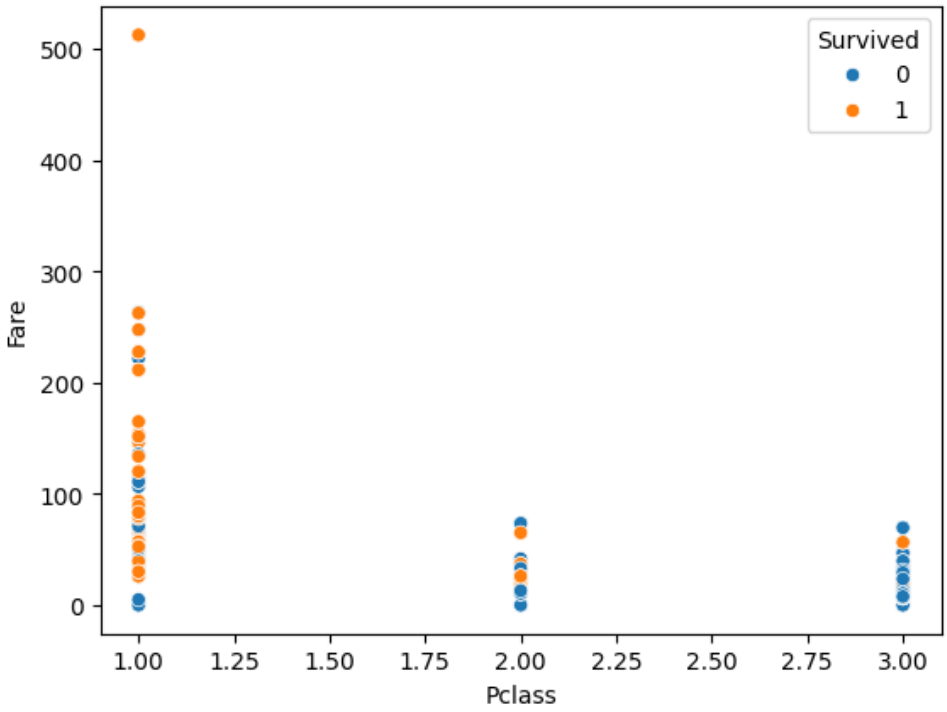
dtype: int64

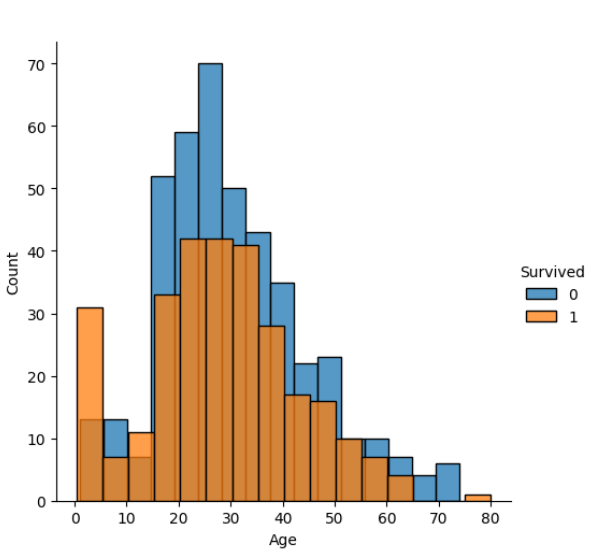


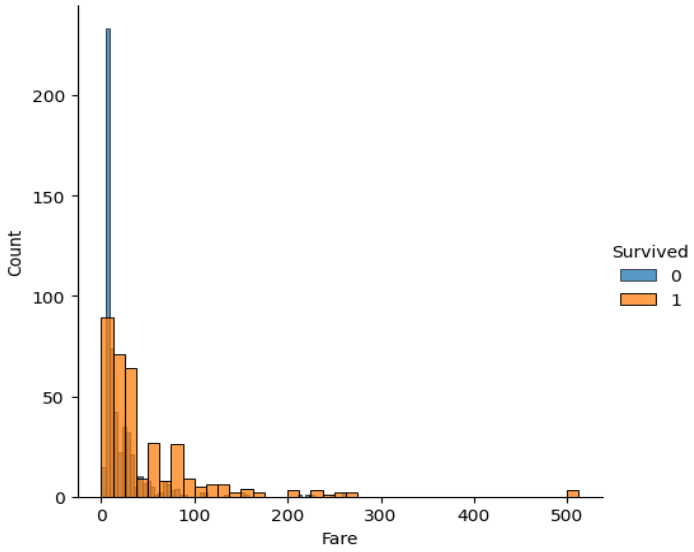
	female	male	PassengerId	Survived	Pclass	Age
0	False	True	1	0	3	22.0
1	True	False	2	1	1	38.0
2	True	False	3	1	3	26.0
3	True	False	4	1	1	35.0
4	False	True	5	0	3	35.0

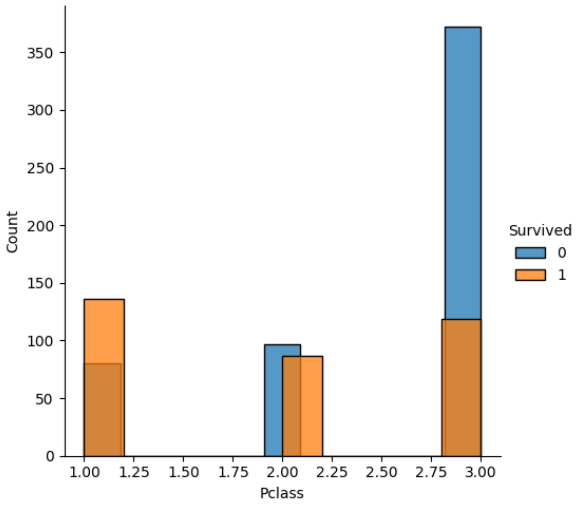


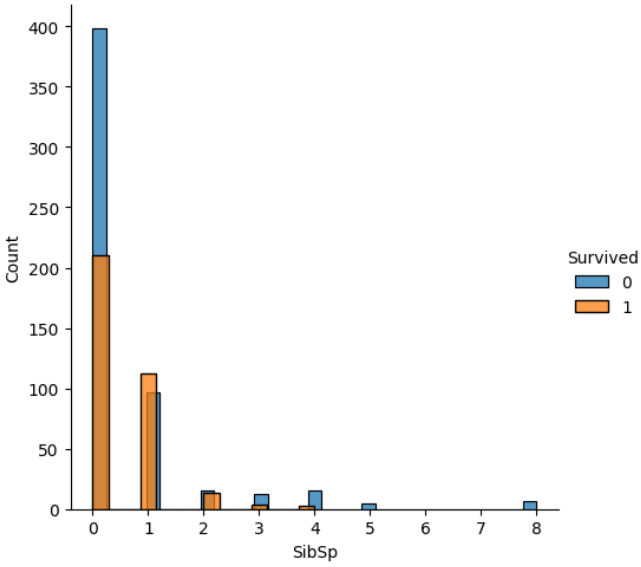

Pclass
Age
SibSp
Parch
Fare
Survived
0
1


Survived
0
1
Fare
Pclass









RESULT:

The Titanic dataset was successfully analyzed using Pandas, Matplotlib, and Seaborn with various statistical analyses and visualizations.